# Grant application full proposal form 2023

## NWO Talent Programme – Vidi scheme

Applied and Engineering Sciences
Social Sciences and Humanities
**Science**

# 1. General information

## 1a. Title of research proposal

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|---|---|
| Title: | Restoring resilient river systems (Resist) |

## 1b. Scientific summary of research proposal

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|---|---|
| Scientific summary: | Europe's ambitious new nature restoration legislation aims to restore 25,000 kilometers of rivers to enhance nature conservation, biodiversity, and climate change mitigation efforts. Resilient river ecosystems with high biodiversity demonstrate reciprocal relationships between physical processes and biotic components, which act as "ecosystem engineers" and are considered Nature-based Solutions. However, little is known about the presence and environmental feedback of ecosystem engineers prior to their eradication or decline in heavily human-modified European rivers, a knowledge gap that significantly hampers river restoration initiatives. This study uses sedimentary ancient DNA (sedaDNA) as a tool for analyzing river sediments, offering unprecedented insights into biota influencing past river processes and vice versa. The approach involves investigating the lateral and vertical transport, deposition, and preservation of sedaDNA within Holocene river sediment stratigraphy, and then validate the findings in several European river reaches currently undergoing process-based river restoration. This project will provide a novel paleoenvironmental context for five iconic river restoration projects, shedding light onto the landscape formation and co-evolution of physical and biotic-driven processes over a Holocene time-scale. Additionally, it will establish a clear roadmap for river practitioners as they design and implement Europe's largest natural restoration scheme. |
| Word count (max. 300 words): | 192 |

## 1c. Keywords

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|---|---|
| Key words (max. 5): | **Fluvial sedimentology, sedaDNA, ecosystem engineering, biodiversity, river restoration** |

## 1d. NWO domain

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|---|---|
| Chosen domain: | Science (ENW) |

## 1e. Main field of research

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|---|---|
| Main field of research: | 50.90.00 Environmental science |
| Other field(s) of research: | 15.30.00 Geodynamics, sedimentation, tectonics, geomorphology |

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|---|---|
| | 49.10.00 Geography |
| | 22.40.00 Ecology |

## 1f. Public summary of research proposal

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|---|---|
| Title: | Beyond dreams of natural streams |
| Public summary: | Europe has a plan to restore 25,000 kilometers of rivers to help nature, protect biodiversity, and fight climate change. Living beings affect how rivers function, but are dependent on these rivers too. Understanding this interaction can help optimize restoration efforts. Today, however, people affect the rivers much more than other living beings. That is why this project wants to learn how these creatures engineered rivers before people had much effect. Using a special kind of DNA found in river dirt, this can be studied to learn how to restore rivers, using this interaction with plants and animals. |
| Word count (50-100 words): | 97 |

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|---|---|
| Title in Dutch: | Niet alleen dromen van natuurlijk stromen |
| Public summary in Dutch: | Europa wil 25.000 kilometer aan rivieren herstellen voor natuurbehoud, biodiversiteit en klimaatverandering. Levende wezens beïnvloeden hoe deze rivieren werken, maar worden zelf ook door de rivieren beïnvloed. Het is belangrijk om deze interactie te begrijpen, zodat het herstel van de rivieren goed uitgevoerd kan worden. Tegenwoordig is het lastig deze interactie te meten omdat mensen veel meer invloed hebben op het systeem dan andere levende wezens. Dit project onderzoekt daarom hoe deze wezens en de rivier elkaar beïnvloedden vóórdat mensen invloed hadden. Dit kan door oud DNA in riviermodder te bestuderen. Met deze kennis kunnen rivieren beter hersteld worden. |
| Word count (50-100 words): | 99 |

# 2. Research proposal

## 2a. Description of proposed research

### 2a1. Overall aim and key objective

### Scientific relevance and challenges

In the face of climate change and fast biodiversity loss society demands actions from governments. The European Union has set out to lead global effort in nature restoration to halt biodiversity loss and mitigate the effects of climate change through creating the globally leading natural protection law, which amongst others includes the challenging goal **to restore 25'000 km of rivers by 2030** (European Commission, 2021). Resilient river ecosystems with high biodiversity exhibit a reciprocal relationships between physical processes and living organisms, which 'engineer' physical processes and are considered 'Nature-based Solutions' (NbS). However, very little is known about the presence and environmental feedbacks associated with **ecosystem engineers** prior to their eradication/decimation in Europe's heavily human modified rivers. This significantly impedes **river restoration** projects, as they are based on assumptions about **natural river dynamics** and ecosystems. Recent advances in the field of **sedimentary ancient DNA (sedaDNA)** gives us the unprecedented opportunity to close this knowledge gap. The method has been highly successful and reliable in other environmental contexts (lake sediments, permafrost, soils), but river sediments have not been utilised due to the river dynamics posing specific challenges to the method. This challenge derives from the fact that river water transports, integrates, and distributes catchment DNA, which means that freshly deposited river sediments likely represent the DNA record of the regional catchment rather than the local conditions, and that the seasonally fluctuating shallow groundwater tables could cause vertical displacement of sedaDNA. **This project aims at validating the sedaDNA record in river-floodplain sediments and develop this method to be able to gain fundamental and new understanding of abiotic-biotic interactions in a past, more natural, palaeoenvironmental precedent of European river-floodplains.** This project will open up new and exciting research into the co-evolution of physical and biotic-driven processes in rivers, and provide a new method to inform river practitioners. The project comprises three main interconnected objectives along three thematic axes which are coupled to specific objectives:

1. **Develop sedimentary ancient DNA (sedaDNA) as a robust tool for river science**
   a. *catchment versus local – sedaDNA record of fresh and subrecent flood deposits*
   b. *groundwater versus adsorption – vertical mobility of sedaDNA in river sediments*

2. **Re-evaluate the role of biotic drivers and ecosystem engineers in the (past) shaping river floodplain dynamics and ecosystems across a European climatic gradient.**
   a. *explain past river dynamics through a detailed, local record of past ecosystem composition using sedaDNA*
   b. *test for the presence of context-specific ecosystem engineers relevant for process-based restoration projects in 5 European climatic zones*

3. **The utility and application of sedimentary DNA (sedaDNA) for river restoration professionals informing process-based river restoration projects**
   a. *develop clear guidelines for the use of sedaDNA in a river restoration context*
   b. *dissemination of relevant information to stakeholders*

**The loss of natural river-floodplain ecosystems.** Thousands of years of human intervention has extensively modified streams, heavily disrupting natural river-floodplain processes (summarized in Brown, 2018). In natural rivers, biota fundamentally influences physical processes and vice versa (Corenblit et al., 2017; Larsen et al., 2021a; Larsen et al., 2016b; Tal and Paola, 2007). The in Europe ubiquitous river engineering has been so effective that it resulted in laterally (river to floodplain) little or disconnected rivers, but vertically highly connected streams ('water and sediment highways'), with serious and long-term implications for riparian ecosystems, water quality, discharge and water residence time all over Europe. In time, engineering has also caused a loss of the understanding of natural riparian systems simply because of their absence (Larsen et al., 2021a). ***It would be fascinating to understand how rivers have functioned before human dominance during paleoenvironmental precedents of intact abiotic-biotic interactions.***

**Paleoenvironmental precedents as baseline for river restoration.** Even though river restoration is a multi-billion industry (Montgomery, 2008), the practice of river restoration has been shown to be often ineffective (summarized in Wohl et al. (2015). Restoration techniques based on natural processes including the strategic use of ecosystem engineering are only reluctantly implemented (Wohl et al., 2024). ***In Europe's heavily modified streams it is imperative***

***to look into the river's historical context to gain information on the natural dynamics, as all human-modified rivers have likely been restricted through engineering from adjusting to their equilibrium state over thousands of years.***

Botanical remains have long been the core of reconstructing past environments, however their fast decomposition under aerobic conditions leads to a focus on rare, non-dynamic, permanently waterlogged locations, e.g. (oxbow) lakes. Furthermore, their origin is often uncertain, as they may have been transported over long distances. ***However, most river restoration projects function, and have the need for information, on a local, river-reach scale.***

**Sedimentary ancient DNA offer significant potential to reveal past ecosystems on a local spatial scale**. SedaDNA is a cutting-edge method in the field of palaeogenetics, allows investigating past ecosystems from sediments in the absence of botanical remains with a high potential for high taxonomic resolution. Several recent simultaneous scientific advances in genetics have created this leap forward, including advances in library generation from double stranded to single stranded, an ever decreasing cost per sequence with advances in high throughput sequence technologies, increasing number of reference genomes of taxonomic assignment, and finally improvements in dedicated programs for bioinformatic analyses of the sequence data. From its first investigation in permafrost soils (Willerslev et al., 2003), sedaDNA has shown to be a new and exciting method for cave and lake sediments, soils, and sediments with an archaeological contexts in many climatic zones to reconstruct past ecosystems, including flora, animals, and humans in unprecedented detail (e.g. Brown et al. (2022), Vernot at al. (2021)). ***Nevertheless, similar to any palaeoecological technique, there is a pronounced necessity to validate the record by conducting contemporary analogue studies to be able to interpret the signal*** (e.g. Edwards et al. (2018); Yoccoz et al. (2012))***, especially in dynamic environments like rivers.***

**Unlocking SedaDNA for river sediments.** Detailed research into the DNA contained and transported in river water (eDNA) provides clear evidence that rivers are conveyor belts of the catchment's DNA, and integrate the catchment's biotic signal (Carraro et al., 2023; Deiner et al., 2016; Mächler et al., 2019). Hence, fresh flood deposition should contain an integrated catchment DNA biotic signal. However, it is clear that soils contain the sedaDNA from the local ecosystem simply because of the large amount of biomass produced which constitutes the source of sed(a)DNA (Yoccoz et al., 2012). ***Therefore, it is imperative to comprehend the timing at which local vegetation supersedes the DNA signal of the catchment in river sediments, in order to effectively utilize sedaDNA for palaeoecological reconstruction.***

In a pilot study at three distinct river reaches we developed a sedaDNA approach which yielded promising results (Figure 1). ***This case study underscores the richness of sedaDNA in river-floodplain deposits and highlights its ability to identify the presence and a much broader diversity of ancient floral species and animal ecosystem engineers compared to traditional palaeoecological methods.***

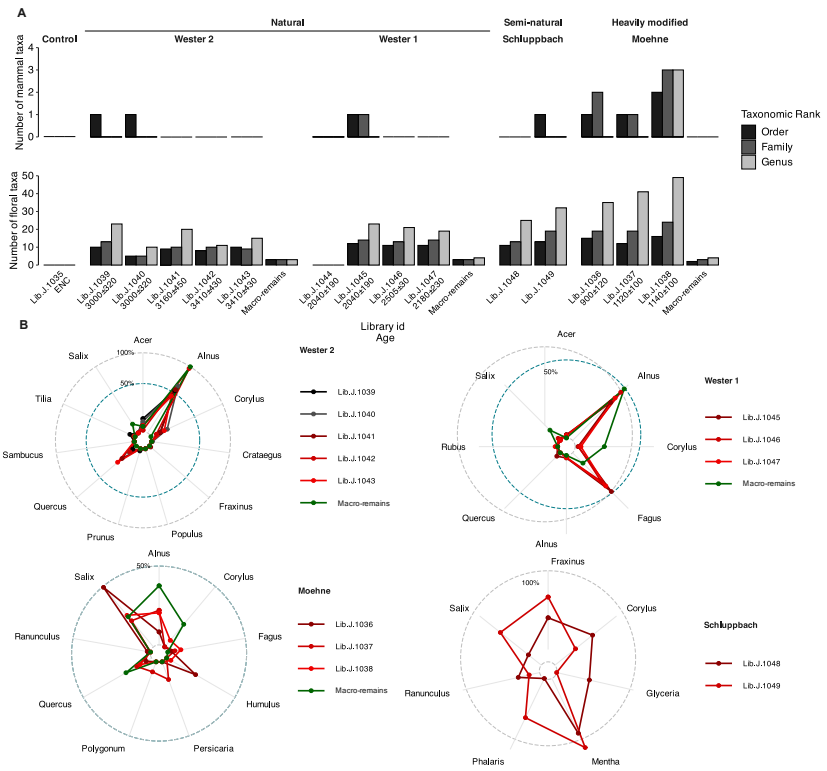


Fig. 1*: Palaeoecological sedaDNA results from three river-floodplains along a human-impact gradient. Panel A) displays single-stranded shotgun sequencing results. The rich sedaDNA record with high complexity indicates that deeper sequencing could reveal even more unique molecules. A high proportion of reads were eukaryotic organisms (0.4-15.2% of unique reads), predominantly from floral species (97-100%). Taxonomic diversity was high in all but one sample. SedaDNA from domestic animals was found in the human-modified site, while the more natural rivers showed few reads assigned to rodents. Additional target capture enriched for human DNA successfully recovered authentic human DNA. Panel (B) displays the six taxa with the highest number of reads from each library, plotted per location alongside macro-remains. All six taxa recorded from macro-remains were also recovered with sedaDNA.*

## Originality and innovative character of the topic

Based on what we know about past river ecosystem change and the behaviour of sedaDNA in river sediments, I hypothesize i) that the sedaDNA record in recent and subrecent deposits represent the catchment, while older river deposits represents a local signal, and ii) that the vertical mobility of DNA within the sedimentary profile depends rather on sediment characteristics than on fluctuating groundwater levels. If this is true, then we will be able to relate the presence of ecosystem engineers functionally to past river dynamics, and advise process-based restoration projects throughout Europe.

In summary, this project will allow us for the first time to take into account important abiotic-biotic feedbacks when reconstructing the quaternary river dynamics, and shed light onto the co-evolution of inhabiting species with physical river-floodplain processes, and how this is key for river restoration projects aiming at nature conservation. This research project is the first attempt to acquire in-depth molecular understanding of past abiotic-biotic interactions in river-floodplains by purposefully integrating genetics and chrono-stratigraphical methods of river-floodplain sediments to disentangle the evolution and ecology of European rivers.

## Research approach of this project

This project applies established, emerging, and novel techniques, which include detailed chrono-stratigraphical methods and molecular techniques such as shotgun sequencing and target capture, in combination with (palaeo)botany, remote sensing, ground-penetrating radar, and GIS-based analysis of available hydrological, geomorphic, land-use and elevation data. Only the new and exciting integration of these approaches is anticipated to shed new light into the past functioning of river systems and their biodiversity. The scope and design of this project will first develop and test the validity of sedaDNA as a method for paleoenvironmental reconstruction from river sediments in one catchment. This refined method will then be implemented in 4 river reaches throughout Europe, subsequently becoming available for use by researchers globally. Each of the six outlined research objectives represent a work package (WP) which produces outcomes that inform the following objective.

Objective 1. Develop sedimentary ancient DNA (sedaDNA) as a robust tool for river science

***WP 1a. Catchment versus local – sedaDNA record of fresh and subrecent flood deposits***

SedaDNA will be extracted from ~100 mg sediment using a protocol optimised for ancient DNA (Rohland et al., 2018). The extracted DNA molecules will be transformed into single-stranded library molecules (Gansauge et al., 2020), that can directly be sequenced by Illumina high-throughput sequencing (shotgun sequencing). After sequencing, the obtained DNA sequences (reads) will be filtered for quality and complexity (reads with highly repetitive sequence) similar to the holi-pipeline (Kjær et al., 2022), before being mapped to a reference database containing nuclear, mitochondrial, and chloroplast DNA from a wide variety of species with Bowtie2. The reads that map to the reference sequence in the reference databases are assigned to the last common ancestor using ngsLCA (Wang et al., 2022). Ancient DNA damage and post-assignment quality assurance are performed by in-house pipeline specifically designed for these samples (Figure 2a).

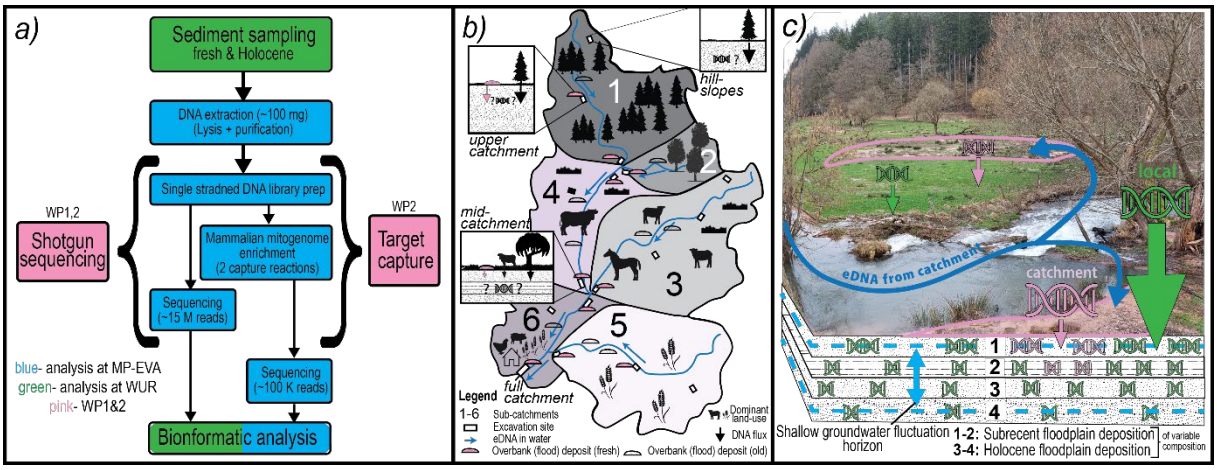


*Fig. 2: Workflow and concepts of WP1. a) Describes the sedaDNA workflow; b) illustrates the spatial distribution of excavation sites using a nested catchment approach for testing upstream-downstream and vertical variability; c) outlines the method for assessing vertical variability of sedaDNA in sedimentary profiles. Arrow size indicates DNA input into sediment layers by different agents, such as fresh deposition (catchment, pink) and local vegetation (green). With depth and DNA degradation, the catchment's sedaDNA signal is expected to diminish. Analysis will also include sedimentary characteristics (layers 1-4) and potential DNA migration due to fluctuating shallow groundwater tables.*

We'll choose a meso-scale catchment (0-4th Strahler stream order) based on criteria like frequent overbank flooding, terrain features, and land use, with sub-catchments showing one dominant land use or vegetation community (Figure 2b). Fresh flood overbank deposits will be analyzed for taxonomic composition using sed(a)DNA, along with density, grain size, organic/inorganic carbon content using RockEval pyrolysis and XRF scanning, potential botanical macro-remains, and pH. Botanical micro-remains have minimal influence on the sedaDNA record (Sjögren et al., 2017). We'll survey surrounding vegetation at each location as per Edwards et al. (2018). Then, we'll create vertical sediment profiles through excavation, and sample flood deposition layers within the first 40 cm for all analyses (subrecent floodplain deposition) (Figure 2c). ***Using a nested catchment approach, we'll expand spatial scale stepwise, investigating the shift from catchment DNA dominance to local DNA dominance, and how this varies with catchment size.***

***WP 1b. Groundwater versus adsorption – vertical mobility of sedaDNA in river sediments***

We will analyze the entire Holocene sedimentary record at all locations, focusing on signs and boundaries of reduction-oxidation (redox) to determine past and present fluctuating shallow groundwater tables (Figure 2c). Excavation sites will be carefully prepared and pumped dry for analysis and sampling, considering groundwater table depth and fluctuations. Sediments will be characterized as in WP1, with additional thin-sections if redox signs are unclear, and deposition age. Sediments will be dated using a mix of Optical Stimulated Luminescence (OSL) and Radiocarbon Dating. SedaDNA samples will be processed as in WP1a and analyzed for intra-site variability. Low variability between samples suggests mixing and fluctuation between sedimentary layers, while distinct taxonomic compositions between layers indicate local deposition with minimal leaching/mixing. ***These results will determine which sedimentary characteristics and environmental conditions control sedaDNA adsorption to sediments, and under what conditions desorption might cause vertical mobility of sedaDNA.***

Objective 2: Re-evaluate the role of biotic drivers and ecosystem engineers in the (past) shaping river floodplain dynamics and ecosystems across a European climatic gradient.

***WP 2a. Explain past river dynamics through a detailed, local record of past ecosystem composition using sedaDNA***

Insights from WP1 on sedaDNA transport, deposition, and preservation in river sediments will inform the strategic selection of four river reaches within iconic restoration sites across Europe, in addition to WP1's research area (Figure 3). Initial sedaDNA findings from WP1, combined with ground penetrating radar and soil corings of each reach, will guide the selection of three excavation sites per reach spanning the entire Holocene sedimentary record and providing a comprehensive understanding of the alluvial architecture (Figure 3). These sites will undergo detailed investigation using sediment characteristics outlined in WP1, with chronology provided by OSL and Radiocarbon Dating. ***A step-wise***

***adaptation of OSL procedures will ensure high quality and resolution. This approach enables the analysis of past river dynamics, considering not only physical processes but also the impact of local ecosystems on these processes and vice versa, for the first time.***

***WP 2b. Test for the presence of context-specific ecosystem engineers relevant for process-based restoration projects***

Shotgun sequencing may not effectively detect taxonomic groups with low biomass, such as ecosystem engineers. Target capture, a molecular method shown to efficiently capture mammalian species in cave sediment (Slon et al., 2017) and permafrost (Murchie et al., 2022) will be employed to enrich for specific ecosystem engineering species relevant for process-based river restoration methods. Target capture, rather than metabarcoding, will allow for the examination of potential intra-species variation using mitochondrial haplotypes. The selection of relevant ecosystem engineers will heavily depend on the local context of the five research areas, for example, eradicated or extinct engineering species like aurochs, bison, beaver, or crayfish (summarized in Rice, 2021). This approach may also include eradicated predators like the wolf, as predator-induced stress can lead to behavioral changes in animal populations altering ecosystem feedbacks (Beschta and Ripple, 2019). ***These results will provide insight into the contributions of long-lost ecosystem engineers to river system dynamics for the first time. They will advise five river restoration projects how to restore using eradicated ecosystem engineers to initiate natural river dynamics creating more sustainable, resilient river ecosystems.***

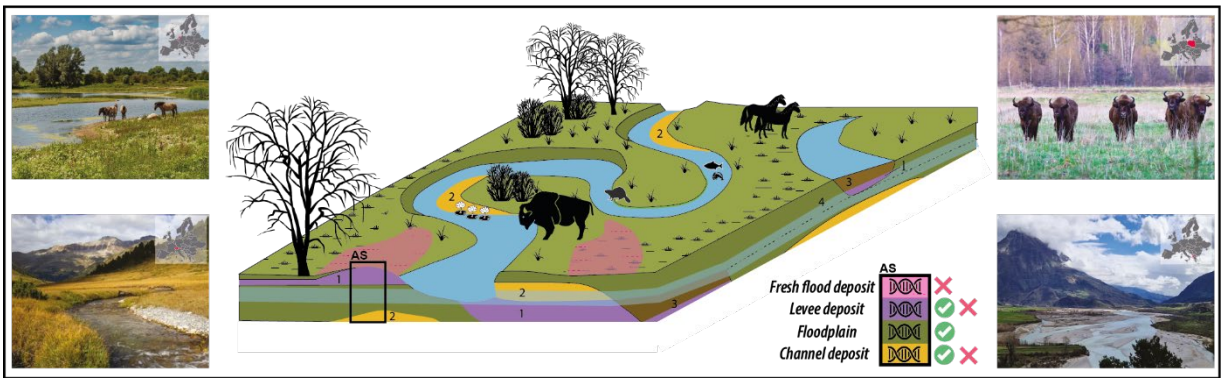


Fig. 3 *illustrates a possible sampling strategy developed based on WP1, for the alluvial stratigraphy (AS) of four iconic rewilded and wild river landscapes in Europe (potential WP2 research areas). The red crosses/green hooks indicate possibly problematic/unproblematic stratigraphic context. The top-left corner is Millingerwaard (NL), top-right corner is Białowieża Forest (PL), bottom-left corner is Swiss National Park (CH), bottom-right corner is the last wild river in Europe, Vjosa (AL). The fifth research area corresponds to WP1. Modified from Larsen et al. (2018).*

Objective 3. The utility and application of sedimentary DNA (sedaDNA) for river restoration professionals informing process-based river restoration projects

***WP 3a. Develop clear guidelines for the use of sedaDNA in a river restoration context***

WP3 builds on results of WP1&2 and focuses on **advising river restoration projects**. Members of this project's user committee (section 3d) and all project members and cooperators will – based on the projects'results - co-develop a concrete guideline for using sedaDNA in restoration projects. This is facilitated through half yearly meetings with the User Committee and a dedicated workshop on the topic (section 2a2). ***This guideline will be published fast and open access and distributed nationally and internationally through stakeholder and WUR networks to ensure visibility.***

***WP 3b. Stakeholder engagement and dissemination of relevant information to stakeholders and policy makers***

Stakeholder engagement is crucial for sharing scientific findings with society. It's integrated into WP1, 2, and 3a, spanning local to international levels. Locally and regionally, stakeholders like waterboards, national park managers, and restoration professionals will influence fieldwork planning and site and engineering species selection. ***All local insight will be compiled into reports, translated, and locally published. Additionally, a policy brief highlighting ecosystem engineering's role in river restoration will be created. Leveraging existing policy networks at the EU level and Wageningen University will ensure knowledge dissemination in the European policy arena.***

## 2a2. Research plan

***Cooperations***. This project requires expertise in river geomorphology, sedimentology, ancient DNA, animal genetics, and ecology. In addition to the applicant's experience in river geomorphology, sedimentology, and ecology, collaboration with three experts is essential: ***Dr. B. Vernot*** from the ***Max Planck Institute for Evolutionary Anthropology (MP-EVA)***, who leads the sedaDNA research group; ***Dr. K. Nota***, responsible for botanical and bioinformatical research within this group; and ***Dr. A. de Groot*** from Wageningen University & Research, an animal geneticist and ecologist, who newly established one of two sedaDNA analysis laboratories in the Netherlands. These partners have complementary expertise and an established collaboration, including shared supervision of PhD candidates. This setup combines cutting-edge facilities and experience from MP with WUR's animal genetics expertise, facilitating knowledge transfer and support for establishing the new laboratory at WUR. A four-day workshop will be organized to foster collaboration among researchers with expertise in sedaDNA, earth-surface-processes, and ecosystem restoration, aiming to produce a benchmark review article. The project is supported by ***Prof. L. Bakker*** (WUR), the only professor in Rewilding Ecology in the Netherlands. Detailed timelines and work packages, along with tangible project outcomes, are outlined below.

| RESist' WORKPLAN | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|
| Research Proposal, Thesis introduction & synthesis | P | P | | T | T |
| WP1 - Unlocking sedaDNA for river sediments | | C S C | C S C | C M S R | |
| WP2 - Investigating the depositional record of 5 wild river-floodplains | | C | C S | C M S R | S C S, C |
| WP3 - clear guidlines for practical use and dissimination | | | | M, R | B C |

*Gantt Chart of Resist. Pink & green: PhD candidate 1 and 2 supervised by applicant and cooperators; blue: Stakeholder engagement, guideline development, publication and dissemination; P: Detailed PhD project research proposal (reviewed by graduate school); S: Peer reviewed scientific paper in reputable international journal; R: Professional publication aiming at water managers, policy makers, restoration practitioners; C: Conference contribution (oral or poster presentation, national and international); M: Meetings with publication goals; B: Blog posts/newsletter/policy briefs/grey publications for practitioners and policy makers; T: PhD thesis.*

***Workplan for requested personnel***. The applicant will lead the research project, and head WP3. Most research activities in WP1 & 2 will be performed by two PhD candidates.

## 2a3. Alignment between research proposal and expertise of the applicant (max. 0.5 page)

My unique research experience, crucial to the proposed project, bridges the study of earth-surface-processes, (geo-)ecology, and archaeology, and is highly collaborative in nature. I strive to take a whole-ecosystem, holistic approach (Figure 4, e.g. Larsen et al., 2021a; Spohn et al., 2021), and work closely with geneticists, biogeochemists, remote sensing/GIS specialists, ecologists, and biologists to advance our understanding of **how abiotic-biotic environmental interactions shape the earth-surface in past, present and future** (Figure 4, Larsen et al., 2021b). To analyse past ecosystem changes, I apply a range of palaeoecological methods (Larsen et al., 2019; Larsen et al., 2016c), including some experience in sedaDNA (Larsen et al., 2018), however my expertise lies in **detailed Holocene chrono-stratigraphy** in variable sedimentary contexts which forms the backbone of this study (Croke et al., 2024; Jankowski et al., 2021; Larsen et al., 2013; Larsen et al., 2016a; Larsen et al., 2016c; Lungershausen et al., 2017; Larsen et al., 2019). The only way of understanding **the effects of past ecosystem engineering is to study them at present,** for which I have used a mix of established and novel monitoring techniques including eDNA, and spatiotemporally upscaled using statistical and physical erosion models (Grigusova et al., 2022; Grigusova et al., 2021; Grigusova et al., 2023; Kraus et al., 2022; Larsen et al., 2016b; Mächler et al., 2021; Robinson et al., 2020). My recent **transdisciplinary research** (Ekberzade et al., 2023a, 2023b, 2024, Larsen, 2023) is influencing nationwide governmental decision making in national riparian corridor management and European policy makers, exemplifying my ability **to work efficiently with stakeholders, managers and national policy makers, which is important to widely disseminate this project's results.** In summary, this cutting-edge project brings my research lines together (Figure 4), and will position my team at the forefront of river science.

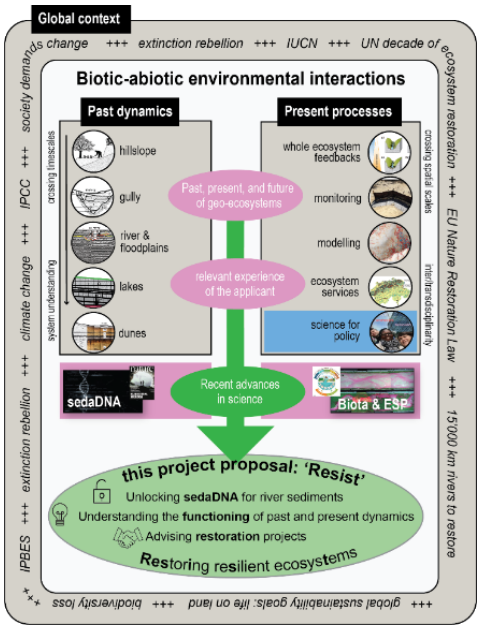


*Fig. 4: Alignment between the applicant's expertise and the project proposal*

### 2a4. Motivation for the choice of host institution

The proposed research will be conducted at Wageningen University & Research's (WUR) Soil Geography and Landscape (SGL) research group. WUR is globally renowned for excellence in Plant and Agricultural Sciences, including Soil Science, offering cutting-edge facilities for soil science and molecular genetics research. My expertise lies in soil science and earth-surface processes, focusing on the interaction between biota and earth-surface processes, which is the unique research expertise I bring to SGL. My team collaborates actively across disciplines, including sedaDNA research, with researchers from the Max Planck Institute for Evolutionary Anthropology (MP-EVA), and Dr. A. de Groot (WUR), who possess crucial complementary expertise and laboratory set-ups. Dr. de Groot leads one of two (sed)aDNA laboratories in the Netherlands, uniquely situated at a university. WUR's SGL group, comprising 7 full-time permanent staff members focusing on river geomorphology offer the expertise and facilities for research on river systems, including river restoration and NbS. WUR also hosts the only institution in the Netherlands employing OSL (NCL) for sediment dating, highly relevant to this proposal. Active collaborations at WUR with plant and animal ecologists will be important to this project. WUR provides top-tier training for PhD candidates and MSc thesis students, supported by tailored supervision, leadership training, and mentoring of the applicant. The university's active policy network and institutional support for public outreach ensure effective knowledge dissemination to the public.

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| Word count (max. 4,000 words): | 3988 |

## 2b. Scientific and/or societal impact of proposed research

**Please do not remove this box, as it contains information for the external referees and the assessment committee.**

The applicant is given the choice to either focus on scientific impact, societal impact, or to focus on both type of impact. One type of impact is not considered superior to the other. The criterion 'Scientific and/or societal impact' can be positively assessed if the applicant only focuses on one type of impact (either scientific or societal) or on both. However, the applicant should clearly motivate their choice and elaborate on the potential impact.
In addition, if the applicant has chosen to primarily focus on one type of impact, the applicant is asked to elaborate how proportional attention to increasing (unforeseen) opportunities for the non-chosen type of impact during the project. Thus, if the focus is on scientific impact, how attention is paid to possible or unforeseen societal impact of the research, and *vice versa*).

| | |
|---|---|
| Chosen focus: | Scientific and societal impact are of comparable focus |

### Motivation

***Scientific impact*.** The study of earth-surface-processes has increasingly considered the role of **biotic factors as controls upon earth-surface-processes** across a wide range of spatial and temporal scales. Where timescales are long (centennial and longer), it has been challenging to quantify relationships between earth-surface-processes and vegetation using botanical remains. Additionally, where the biotic agents are fauna, longer term reconstruction of the impacts of biological activity upon geomorphic processes has been mostly impossible. This project aims at developing sedaDNA as a molecular proxy to decipher the presence and nature of faunal influences on earth-surface-processes in fluvial deposits on a local scale. ***When used appropriately, this method has the potential to greatly improve our understanding of biotic drivers of river dynamics over long timescales, and so to reconstruct how such drivers might explain the landscape as we see it today*** (Larsen et al., 2018).

**Alluvial river systems** in Europe have been argued to be originally of a multi-threat, anastomosing, and wooded nature (summarized in Brown et al., 2018), however there is only a handful of studies, which prohibit extrapolation to a larger spatial scale. Due to human impact, most European river systems have transformed to the at present dominant single channel systems characterised by sinuous or straight channels (Brown et al., 2018). Much research has been done on anthropogenically modified rivers, but not into the natural Holocene dynamics, which is likely because of the thin, discontinuous and heterogeneous nature of these deposits, and their low position close to the shallow groundwater table (Walter and Merritts, 2008). ***In this project, we aim at greatly advancing the field by creating a toolbox to***

***research the natural dynamics of streams were prior to human dominance, which will greatly enhance our understanding of the European landscape and its transformation in the Anthropocene***.

Validating the local origin of **sedaDNA** in river sediments opens up another underutilized sediment type for the scientific community. ***The foundational work of this project paves the way for researchers to recognize the potential of river deposits in studying past ecosystems. Depending on the sample quality, there is potential to harness population/individual level genetic information, which represents a new frontier in the sedaDNA community.***

***Societal impact***. The EU's **new nature restoration law** will initiate the restoration of 25,000 km of European streams using nature-based solutions (European Commission, 2021). This legislation aligns with the global call from the **IUCN for a decade of ecosystem restoration** to address issues like deforestation, land degradation, pollution, and climate change, as well as to reverse biodiversity loss and decline in ecosystem services (IUCN, 2021). These actions are part of a global initiative aimed at achieving significant progress towards the **Global Sustainability Goals** (United Nations, 2015a).

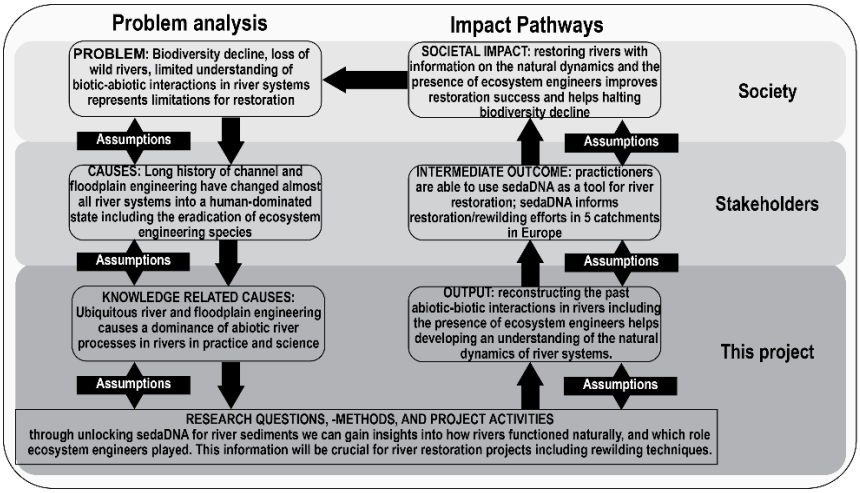


*Fig. 5: Theory of Change (Resist)*

This project aims to **enhance river restoration practices towards more natural and sustainable river management**, allowing for river dynamics to be moderated by ecosystem engineers (Figure 5). It is timely amidst the global push for restoration efforts in Europe and beyond. By developing the sedaDNA method for rivers and their restoration, we aim not only to inform 5 ongoing river restoration projects in Europe, but also to provide a blueprint for using the method to improve river restoration worldwide. This approach aims at contributing to global efforts to achieve ***Sustainable Development Goal 15 (Life on Land)*** by protecting, restoring, and promoting the sustainable use of riparian ecosystems and halting biodiversity loss (Figure 5) (United Nations, 2015a). Additionally, it supports efforts to combat climate change and its impacts (***Sustainable Development Goal 13***) by providing information for more sustainable river management practices (Figure 5) (United Nations, 2015a, United Nations, 2015b).

We will **engage a diverse array of stakeholders** from both the public and private sectors, including practitioners, governmental and non-governmental organizations, as well as citizens, alongside a network with policymakers, which will facilitate the **dissemination of scientific findings widely into practise, society and policy**. Through active involvement of the User Committee and stakeholders throughout the duration of the project (section 2a2), our objective is to collaboratively develop guidelines for the application of sedaDNA in river restoration. This process will culminate in a workshop where the project's findings will be summarized and tailored specifically for the river restoration practitioner community. Furthermore, the project's outcomes will be integrated into a policy brief aimed at informing national and international policymakers about ecosystem engineering as NbS. On a local level, this project specifically aims at adopting an open, transdisciplinary science approach. We aim to integrate local knowledge and perspectives into our research decisions, incorporating the interests and insights of the local community. Reports about the context-specific results of each research area will be formulated targeted towards the public and in the local language, which builds local support for science and ecosystem history, and over the long-term may lead to more appreciation of the value of ecosystems and their protection (Andersen, 2022).

| | |
|---|---|
| Word count (max. 1,000 words): | 994 |

## 2c. Literature references


Andersen, I., 2022. Science-based Solutions for a Sustainable Planet, United Nation Speeches. United Nations, Stockholm.

Beschta, R.L. and Ripple, W.J., 2019. Can large carnivores change streams via a trophic cascade? Ecohydrology, 12(1): e2048.

Brown, A.G., Fonville, T., van Hardenbroek, M., Cavers, G., Crone, A., McCormick, F., Murray, E., Mackay, H., Whitehouse, N.J., Henderson, A.C.G., Barratt, P., Davies, K., Head, K., Langdon, P., Alsos, I.G. and Pirrie, D., 2022. New integrated molecular approaches for investigating lake settlements in north-western Europe. Antiquity, 96(389): 1179-1199.

Carraro, L., Blackman, R.C. and Altermatt, F., 2023. Modelling environmental DNA transport in rivers reveals highly resolved spatio-temporal biodiversity patterns. Scientific Reports, 13(1): 8854.

Corenblit, D., Steiger, J. and Till-Bottraud, I., 2017. Biogeomorphic feedbacks within riparian corridors: the role of positive interactions between riparian plants, EGU General Assembly Conference Abstracts, pp. 8457.

Croke, J., Thompson, C., Larsen, A., Macklin, M. and Hughes, K., 2024. Fluvial Response to Environmental Change in Sub-Tropical Australia over the Past 220 Ka, Quaternary.

Deiner, K., Fronhofer, E.A., Mächler, E., Walser, J.-C. and Altermatt, F., 2016. Environmental DNA reveals that rivers are conveyer belts of biodiversity information. Nature Communications, 7: 12544.

Edwards, M.E., Alsos, I.G., Yoccoz, N., Coissac, E., Goslar, T., Gielly, L., Haile, J., Langdon, C.T., Tribsch, A., Binney, H.A., von Stedingk, H. and Taberlet, P., 2018. Metabarcoding of modern soil DNA gives a highly local vegetation signal in Svalbard tundra. The Holocene, 28(12): 2006-2016.

Ekberzade, B., Carrasco, A.R., Izdebski, A., Sofo, A., Larsen, A., Akinyemi, F.O., Bruckman, V.J., Baker, N., Clark, S. and Hill, C., 2024. GC Insights: Fostering transformative change for biodiversity restoration through transdisciplinary research. Geosci. Commun., 7(1): 57-61.

Ekberzade, B.C., Rita; Polo, María J.; Bruckman, Viktor J.; Sofo, Adriano; Izdebski, Adam; Akinyemi, Felicia; Baker, Noel; Clark, Simon; Larsen, Annegret, 2023a. Restoring forest ecosystems: Evidence-based insights for policymakers.

Ekberzade, B.C., Rita; Polo, María J.; Bruckman, Viktor J.; Sofo, Adriano; Izdebski, Adam; Akinyemi, Felicia; Baker, Noel; Clark, Simon; Larsen, Annegret, 2023b. Restoring peatlands: Evidence-based insights for policymakers. European Geosciences Union.

European Commission, 2021. Biodiversity Strategy 2030 - Barrier Removal for River Restoration. doi:10.2779/181512.

Gansauge, M.-T., Aximu-Petri, A., Nagel, S. and Meyer, M., 2020. Manual and automated preparation of single-stranded DNA libraries for the sequencing of DNA from ancient biological remains and other sources of highly degraded DNA. Nature Protocols, 15(8): 2279-2300.

Grigusova, P., Larsen, A., Achilles, S., Brandl, R., del Río, C., Farwig, N., Kraus, D., Paulino, L., Pliscoff, P., Übernickel, K. and Bendix, J., 2022. Higher sediment redistribution rates related to burrowing animals than previously assumed as revealed by time-of-flight-based monitoring. Earth Surf. Dynam., 10(6): 1273-1301.

Grigusova, P., Larsen, A., Achilles, S., Klug, A., Fischer, R., Kraus, D., Übernickel, K., Paulino, L., Pliscoff, P., Brandl, R., Farwig, N. and Bendix, J., 2021. Area-Wide Prediction of Vertebrate and Invertebrate Hole Density and Depth across a Climate Gradient in Chile Based on UAV and Machine Learning, Drones.

Grigusova, P., Larsen, A., Brandl, R., del Río, C., Farwig, N., Kraus, D., Paulino, L., Pliscoff, P. and Bendix, J., 2023. Mammalian bioturbation amplifies rates of both hillslope sediment erosion and accumulation along the Chilean climate gradient. Biogeosciences, 20(15): 3367-3394.

International Union for the Conservation of Nature, 2021. Decade of Ecosystem Restoration: Launching the Global Movement, https://www.decadeonrestoration.org/.

Jankowski, N.R., Cohen, T.J., Larsen, J., Larsen, A. and May, J.H., 2021. Revisiting the abandoned shorelines of Lake George, Australia: a refined optical dating framework. Journal of Quaternary Science, 36(6): 1052-1072.

Kjær, K.H., Winther Pedersen, M., De Sanctis, B., De Cahsan, B., Korneliussen, T.S., Michelsen, C.S., Sand, K.K., Jelavić, S., Ruter, A.H., Schmidt, A.M.A., Kjeldsen, K.K., Tesakov, A.S., Snowball, I., Gosse, J.C., Alsos, I.G., Wang, Y., Dockter, C., Rasmussen, M., Jørgensen, M.E., Skadhauge, B., Prohaska, A., Kristensen, J.Å., Bjerager, M., Allentoft, M.E., Coissac, E., Alsos, I.G., Coissac, E., Rouillard, A., Simakova, A., Fernandez-Guerra, A., Bowler, C., Macias-Fauria, M., Vinner, L., Welch, J.J., Hidy, A.J., Sikora, M., Collins, M.J., Durbin, R., Larsen, N.K., Willerslev, E. and PhyloNorway, C., 2022. A 2-million-year-old ecosystem in Greenland uncovered by environmental DNA. Nature, 612(7939): 283-291.

Kraus, D., Brandl, R., Achilles, S., Bendix, J., Grigusova, P., Larsen, A., Pliscoff, P., Übernickel, K. and Farwig, N., 2022. Vegetation and vertebrate abundance as drivers of bioturbation patterns along a climate gradient. PLOS ONE, 17(3): e0264408.

Larsen, A., Alvarez, N., Sperisen, C. and Lane, S.N., 2018. Biotic drivers of river and floodplain geomorphology – New molecular methods for assessing present-day and past biota. Earth Surface Processes and Landforms, 43(1): 333-338.

Larsen, A., Bork, H.-R., Fuelling, A., Fuchs, M. and Larsen, J.R., 2013. The processes and timing of sediment delivery from headwaters to the trunk stream of a Central European mountain gully catchment. Geomorphology, 201(0): 215-226.

Larsen, A., Heckmann, T., Larsen, J.R. and Bork, H.-R., 2016a. Gully catchments as a sediment sink, not just a source: Results from a long-term (~12 500 year) sediment budget. Earth Surface Processes and Landforms, 41(4): 486-498.

Larsen, A., Larsen, J.R. and Lane, S.N., 2021a. Dam builders and their works: Beaver influences on the structure and function of river corridor hydrology, geomorphology, biogeochemistry and ecosystems. Earth-Science Reviews, 218: 103623.

Larsen, A., May, J.-H. and Carah, X., 2019. Late Quaternary biotic and abiotic controls on long-term sediment flux in a northern Australian tropical river system. Earth Surface Processes and Landforms, 44(12): 2494-2509.

Larsen, A., May, J.-H., Moss, P. and Hacker, J., 2016b. Could alluvial knickpoint retreat rather than fire drive the loss of alluvial wet monsoon forest, tropical northern Australia? Earth Surface Processes and Landforms, 41(11): 1583-1594.

Larsen, A., Nardin, W., van de Lageweg, W.I. and Bätz, N., 2021b. Biogeomorphology, quo vadis? On processes, time, and space in biogeomorphology. Earth Surface Processes and Landforms, 46(1): 12-23.

Larsen, A., Robin, V., Heckmann, T., Fülling, A., Larsen, J.R. and Bork, H.-R., 2016c. The influence of historic land-use changes on hillslope erosion and sediment redistribution. The Holocene.

Larsen, A., Dennis, M.; Angst, Christof, 2023. Spatial model of beaver impact. In: Federal Office of the Environment (CH) (Editor), https://vdc.wsl.ch/vdc_expert.

Lungershausen, U., Larsen, A., Bork, H.-R. and Duttmann, R., 2017. Anthropogenic influence on rates of aeolian dune activity within the northern European Sand Belt and socio-economic feedbacks over the last ~2500 years. The Holocene: 0959683617715693.

Mächler, E., Salyani, A., Walser, J.C., Larsen, A., Schaefli, B., Altermatt, F. and Ceperley, N., 2019. Water tracing with environmental DNA in a high-Alpine catchment. Hydrol. Earth Syst. Sci. Discuss., 2019: 1-30.

Mächler, E., Salyani, A., Walser, J.C., Larsen, A., Schaefli, B., Altermatt, F. and Ceperley, N., 2021. Environmental DNA simultaneously informs hydrological and biodiversity characterization of an Alpine catchment. Hydrol. Earth Syst. Sci., 25(2): 735-753.

Montgomery, D.R., 2008. Dreams of Natural Streams. Science, 319(5861): 291-292.

Murchie, T.J., Karpinski, E., Eaton, K., Duggan, A.T., Baleka, S., Zazula, G., MacPhee, R.D.E., Froese, D. and Poinar, H.N., 2022. Pleistocene mitogenomes reconstructed from the environmental DNA of permafrost sediments. Current Biology, 32(4): 851-860.e7.

Pedersen, M.W., Ruter, A., Schweger, C., Friebe, H., Staff, R.A., Kjeldsen, K.K., Mendoza, M.L.Z., Beaudoin, A.B., Zutter, C., Larsen, N.K., Potter, B.A., Nielsen, R., Rainville, R.A., Orlando, L., Meltzer, D.J., Kjær, K.H. and Willerslev, E., 2016. Postglacial viability and colonization in North America's ice-free corridor. Nature, 537(7618): 45-49.

Rice, S.P., 2021. Why so skeptical? The role of animals in fluvial geomorphology. WIREs Water, 8(6): e1549.

Robinson, C.T., Schweizer, P., Larsen, A., Schubert, C.J. and Siebers, A.R., 2020. Beaver effects on macroinvertebrate assemblages in two streams with contrasting morphology. Science of The Total Environment, 722: 137899.

Rohland, N., Glocke, I., Aximu-Petri, A. and Meyer, M., 2018. Extraction of highly degraded DNA from ancient bones, teeth and sediments for high-throughput sequencing. Nature Protocols, 13(11): 2447-2461.

Shogren, A.J., Tank, J.L., Andruszkiewicz, E., Olds, B., Mahon, A.R., Jerde, C.L. and Bolster, D., 2017. Controls on eDNA movement in streams: Transport, Retention, and Resuspension. Scientific Reports, 7(1): 5065.

Slon, V., Hopfe, C., Weiß, C.L., Mafessoni, F., de la Rasilla, M., Lalueza-Fox, C., Rosas, A., Soressi, M., Knul, M.V., Miller, R., Stewart, J.R., Derevianko, A.P., Jacobs, Z., Li, B., Roberts, R.G., Shunkov, M.V., de Lumley, H., Perrenoud, C., Gušić, I., Kućan, Ž., Rudan, P., Aximu-Petri, A., Essel, E., Nagel, S., Nickel, B., Schmidt, A., Prüfer, K., Kelso, J., Burbano, H.A., Pääbo, S. and Meyer, M., 2017. Neandertal and Denisovan DNA from Pleistocene sediments. Science, 356(6338): 605-608.

Spohn, M., Aburto, F., Ehlers, T.A., Farwig, N., Frings, P.J., Hartmann, H., Hoffmann, T., Larsen, A. and Oelmann, Y., 2021. Terrestrial ecosystems buffer inputs through storage and recycling of elements. Biogeochemistry, 156(3): 351-373.

Tal, M. and Paola, C., 2007. Dynamic single-thread channels maintained by the interaction of flow and vegetation. Geology, 35(4): 347-350.

United Nations, 2015a. The 2030 Agenda for Sustainable Development. https://documents.un.org/doc/undoc/gen/n15/291/89/pdf/n1529189.pdf?token=g3ZdmxLjbgpA75r8Bk&fe=true, New York, pp. 35.

United Nations, 2015b. Adoption of the Paris Agreement. https://documents.un.org/doc/undoc/ltd/g15/283/19/pdf/g1528319.pdf?token=mYAURXksMYsvsZj9fb&fe=true.

Vernot, B., Zavala, E.I., Gómez-Olivencia, A., Jacobs, Z., Slon, V., Mafessoni, F., Romagné, F., Pearson, A., Petr, M., Sala, N., Pablos, A., Aranburu, A., de Castro, J.M.B., Carbonell, E., Li, B., Krajcarz, M.T., Krivoshapkin, A.I., Kolobova, K.A., Kozlikin, M.B., Shunkov, M.V., Derevianko, A.P., Viola, B., Grote, S., Essel, E., Herráez, D.L., Nagel, S., Nickel, B., Richter, J., Schmidt, A., Peter, B., Kelso, J., Roberts, R.G., Arsuaga, J.-L. and Meyer, M., 2021. Unearthing Neanderthal population history using nuclear and mitochondrial DNA from cave sediments. Science, 372(6542): eabf1667.

Wang, Y., Korneliussen, T.S., Holman, L.E., Manica, A. and Pedersen, M.W., 2022. ngsLCA—A toolkit for fast and flexible lowest common ancestor inference and taxonomic profiling of metagenomic data. Methods in Ecology and Evolution, 13(12): 2699-2708.

Willerslev, E., Hansen, A.J., Binladen, J., Brand, T.B., Gilbert, M.T.P., Shapiro, B., Bunce, M., Wiuf, C., Gilichinsky, D.A. and Cooper, A., 2003. Diverse Plant and Animal Genetic Records from Holocene and Pleistocene Sediments. Science, 300(5620): 791-795.

Wohl, E., Lane, S.N. and Wilcox, A.C., 2015. The science and practice of river restoration. Water Resources Research, 51(8): 5974-5997.

Wolf, E.C., Cooper, D.J. and Hobbs, N.T., 2007. Hydrologic regime and herbivory stabilize an alternative state in Yellowstone National Park. Ecological Applications, 17(6): 1572-1587.

Yoccoz, N.G., BrÅThen, K.A., Gielly, L., Haile, J., Edwards, M.E., Goslar, T., Von Stedingk, H., Brysting, A.K., Coissac, E., Pompanon, F., SØNstebØ, J.H., Miquel, C., Valentini, A., De Bello, F., Chave, J., Thuiller, W., Wincker, P., Cruaud, C., Gavory, F., Rasmussen, M., Gilbert, M.T.P., Orlando, L., Brochmann, C., Willerslev, E. and Taberlet, P., 2012. DNA from soil mirrors plant taxonomic and growth form diversity. Molecular Ecology, 21(15): 3647-3655.

## 2d. Data management

1. Will this project involve reusing existing research data?

☒ Yes: Are there any constraints on its reuse? Why?

☐ No: Have you considered reusing existing data, but discarded the possibility? Why?

| | |
|---|---|
| Explanation: | The reference genomes/sequences which are used are fully available on public genetic repositories such as GenBank and ENA. The project will also re-use data collected from a previous pilot project of the applicant and existing ecological/land-use/surface terrain data with no constraints attached. |

2. Will data be collected or generated that are suitable for reuse?

☒ Yes: *Please answer questions 3 and 4.*

☐ No: Please explain why the research will not result in reusable data, or in data that cannot be stored, or data that for other reasons are not relevant for reuse. *You do not have to answer questions 3 and 4 and you can remove them.*

| | |
|---|---|
| Explanation (if 'no'): | Explanation |

3. After the project has been completed, how will the data be stored for the long-term and made available for reuse by third parties? Are there possible restrictions to data sharing or embargo reasons? Please elaborate.

| | |
|---|---|
| Explanation: | **SedaDNA data and results**: The raw sequence data will be deposited on the NCBI Sequence Read Archive (SRA), or ENA. There are no reasons for embargoes or restrictions for making the data available. **Geo-samples and results**: Geo-sample metadata and analysis results will be added free of charge to a Geodataserver for geological samples (e.g. Geodataserver GFZ, https://dataservices.gfz-potsdam.de/portal/about.html) and obtain a unique identifier (IGSN) with which all data related to the samples can be accessed without constraints. There are no reasons for restrictions or embargos for making data available. |

4. Will any costs (financial and time) related to store and prepare data for sharing/preservation be incurred?

☐ Yes: *Please be sure to specify the associated expenses in the budget table of this application form.*

x No: All the necessary resources (financial and time) to store and prepare sediment samples and data for sharing/preservation are or will be available at no extra cost. There is no cost associated with depositing raw sequencing data and sediment data.

# 3. Budget

## 3a. Budget table

| *Overview budget* | Description | FTE | Months | Amount |
|---|---|---|---|---|
| **1. Personnel** | | | | |
| *Scientific personnel* | Applicant A. Larsen | 0,5 | 60 | 278.281 |
| *Scientific personnel* | PhD 1 | 1 | 48 | 247.057 |
| *Non-scientific personnel* | PhD 2 | 1 | 48 | 247.057 |
| **Total personnel** | | | | 772.395 |
| **2. Material** | | | | |
| | Lab Analysis | | | 229.800 |
| | Conferences | | | 16.500 |
| | Fieldwork | | | 20.000 |
| | Material costs | | | 6.000 |
| **Total material** | | | | 272.300 |
| **3. Knowledge utilisation** | | | | |
| | Workshop knowledge dissemination | | | 2.500 |
| **Total knowledge utilisation** | | | | 2.500 |
| **4. Internationalisation** | | | | |
| | sedaDNA writing retreat/workshop | | | 5.400 |
| **Total internationalisation** | | | | 5.400 |
| **Grand total** | | | | **1.052.595** |

## 3b. Contribution by the host institution

| *Contribution by the host institution* | |
|---|---|
| Description | Value (in euros) |
| Personnel costs | 202.595 |

## 3c. Co-funding by a third party

| *Co-funding* | | | |
|---|---|---|---|
| Stakeholder | Description | Value (in euros) | Contribution (in-kind/in-cash) |
| n/a | Description | Value in € | Choose in-kind/in-cash |
| n/a | Description | Value in € | Choose in-kind/in-cash |

## 3d. Interest/benefits for stakeholders (if applicable)

| | |
|---|---|
| Will there be any stakeholders involved in a user committee in the proposed project? If so, please describe the involved stakeholders and briefly motivate their interest and involvement in the proposed project in the next table.<br>***Please note: for applications within the AES domain at least 4 stakeholders should be mentioned.*** | Yes: please fill out the table below |

| Name and/or logo for each stakeholder: | Interest and benefits for stakeholders in user committee:<br>*Please do not use more than 70 words on average for the description of each user.* |
|---|---|

| | |
|---|---|
| STOWA (Stichting Toegepast Onderzoek Waterbeheer) | The project proposal aligns perfectly with STOWA's approach to restore rivers based on the natural processes and create resilient ecosystems along them, while at the same time secure water safety in the Netherlands. |
| European Association for River Restoration | In the coming decade of restoration, enhanced understanding of using nature-based solutions to restore rivers is vital. Success in this project will empower river restoration practitioners to utilize the proposed method for a clearer comprehension of natural conditions, serving as a baseline for current restoration efforts. |
| ARK – Rewilding Nederland | If successful, the project will contribute crucial data on where to reintroduce specific species and their habitat needs. This scientific information will be essential for decision-making and managing conservation sites effectively. |
| Zoogdiervereiniging | The project's findings will provide crucial insights into past animal presence and their natural habitats, addressing a current gap in wildlife management. |

## 3e. Totals

| | |
|---|---|
| Grand total: | 1052595 |
| Contribution by the host institution: | 202595 |
| Co-funding by a third party: | 0 |
| **Total requested Vidi-budget:** | **850000** |

## 3f. Additional (applications for) funding for overlapping project(s)

| | |
|---|---|
| Have you applied for and/or obtained any additional funding for (parts of) this project either from NWO or from any other institution, and/or has the same idea been submitted elsewhere? | No |

| | |
|---|---|
| Explanation (if 'yes') (max. 100 words): | Explanation, if applicable |

# 4. Administrative details

## 4a. Ethical aspects

| | |
|---|---|
| Approval from a medical ethics review committee: | Not applicable |
| Approval from an animal experiments committee: | Not applicable |
| Permission for research with the population screening Act: | Not applicable |
| Approval from any other recognised ethics review committee(s): | Not applicable |

## 4b. Closing statement

| | |
|---|---|
| I have included a 'Statement exceeding maximum funding': | Yes |
| I have included a 'Declaration co-funding': | No |
| By submitting this form, I declare that I have completed this application form truthfully, and that I satisfy the nationally and internationally accepted standards for scientific conduct as stated in the 'Netherlands Code of Conduct for Research Integrity 2018 (https://www.nwo.nl/en/news/new-netherlands-code-conduct-research-integrity). I endorse the Code of Conduct for Laboratory Animals and the Code of Conduct for Biosecurity/possibility for dual-use of the expected results and will act accordingly, if applicable. | |

| | |
|---|---|
| Title(s), initial(s) and last name: | Dr. A. Larsen |
| Preferred language of correspondence: | English |
| Place: | Wageningen |
| Date: | 26/3/2024 |